

3. Strategies for Improving GroScore.

3.1. Introduction

In the previous chapter, a novel method for the scoring of many docked protein-protein complexes using atomistic MD simulations with explicit solvent was presented. The approach using free-energy calculations with a robust statistical-mechanical foundation was able to find high-quality structures in many benchmark sets. However, for some targets, the scoring pipeline failed to score near-native poses in high ranks, while other previously described methods were able to discriminate near-native poses from non-native poses. This was especially apparent for the CAPRI target sets T89 and T97,¹ where the best scorers, Oliva² and Zou,^{3, 4} respectively, were able to rank medium-quality structures in their Top-10 ranked poses, whereas the GroScore method could not place any near-native poses within the Top-10 ranked complexes.

In this chapter, two improvements to the GroScore scoring pipeline will be introduced and discussed. First, the implementation of a bidirectional pulling approach (i.e., the simulation of both protein unbinding and binding to obtain binding works) will be described. Based on previous work on the efficiency of bidirectional non-equilibrium simulations,⁵⁻⁹ it could be expected that a bidirectional approach in which both unbinding and binding works are combined to a single binding-free energy estimate offers a possibility for more efficient scoring. Second, given the often non-globular shape of investigated protein structures, a method was developed in which only protein atoms within a certain radius of the interface regions are considered for simulation, leading to much smaller simulation boxes and hence savings in computational resources for some targets.

Both approaches were implemented into the GroScore scoring pipeline and tested with three selected CAPRI scoring sets, targets T41, T47 and T89.^{2, 10} This subset of benchmark sets was selected as the performance of the original GroScore pipeline was very different for these three sets: For T47, many high-quality structures were selected, whereas for T41, only medium-quality poses were ranked within the Top-10, despite high-quality poses being part of the benchmark set. For T89, no near-native poses were found.

3.2. Methods

The bidirectional pulling and interface-only simulation functionalities were implemented into the automated GroScore pipeline interacting with GROMACS 2019.5.¹¹ Other functionalities of the GroScore protocol were kept as previously described. In short, the protein complex structures were checked for entanglements between the binding partners and chain-breaks near the interface region and were subsequently parametrized using the GROMOS 54A8 force-field.^{12, 13} Structures were energy minimized in vacuo, solvated, and equilibrated to a reference temperature of 310 K in an NPT ensemble at a reference pressure of 1 bar. For a detailed description of used parameters and methods the reader is referred to the previous literature.¹

3.2.1. Bidirectional Non-Equilibrium Simulation

For simulation of bidirectional binding, the previously employed method of defining distance restraints between the binding partners using only the C α atoms of the interface residues was not practicable, as the short time period of the non-equilibrium simulation of protein binding led to incorrect sidechain packing in the interface, resulting in large positive binding works ΔW_{bind} (data

not shown). Instead, distance restraints were defined using all non-hydrogen atoms. All possible vectors between non-hydrogen atoms connecting the binding partners in the interface region up to a distance of 0.6 nm were used as distance restraints. The other parametrizations of the distance restraints were done as previously described: The sum of the equally partitioned force constants of the single distance restraints is $2.5 \times 10^4 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{nm}^{-2}$; i.e., the force constants of the single distance restraints were set to $k_i = \frac{2.5 \times 10^4 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{nm}^{-2}}{n_{\text{restraints}}}$. Nonequilibrium unbinding was simulated for 5 ns by increasing the restrained distances by $0.0002 \text{ nm}\cdot\text{ps}^{-1}$, i.e., by up to 1.0 nm. The nonequilibrium work needed for the separation of the binding partners ΔW_{unbind} was obtained from the sum of the integrated forces of the single distance restraints.¹ After unbinding, the proteins were shortly simulated in equilibrium for 100 ps, before binding was simulated in the opposite direction for 5 ns by decreasing the restrained distances by $0.0002 \text{ nm}\cdot\text{ps}^{-1}$ to obtain ΔW_{bind} in the same way (see **Figure 3-1** below).

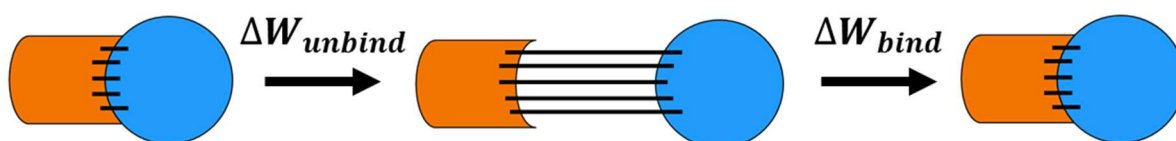


Figure 3-1. Unbinding of a single docked pose of two binding partners (orange and blue proteins) is performed by increasing the restrained distance of many intermolecular distance restraints (black lines) over the time of 5 ns¹ to obtain ΔW_{unbind} . Similarly, binding is simulated by decreasing the restrained distance to obtain ΔW_{bind} .

Unbinding and (re-)binding was simulated once per pose, leading to the same total 2-fold simulation coverage achieved with the previous unidirectional simulation protocol.¹ The obtained single works were combined to a binding free energy estimate (GroScore) by taking the arithmetic mean of the negative ΔW_{unbind} and ΔW_{bind} .

3.2.2. Simulation of Only the Protein-Protein Interface Region (Trimming)

To decrease the size of the used simulation boxes and thereby the computational resources needed for scoring, the additional option to only simulate protein atoms in a certain radius around the scored protein-protein interface was added. For this, a certain radius from the protein-protein interface (the protein-protein interface was defined as all atoms within 0.6 nm of the other binding partner) was defined beyond which all protein residues were removed from the simulated structure ("trimming cut-off"). Three different radii for trimming were tested: 1.5 nm, 2.0 nm and 2.5 nm. To keep the remaining protein structures from falling apart during the simulation, introduced artificial chain-breaks were identified and C α -atoms of the artificial termini restrained by an elastic network approach:¹⁴⁻¹⁷ distance restraints were defined to C α -atoms of other protein residues within the same original chain within a lower cut-off of 0.4 nm and a higher cut-off of 0.9 nm and a force constant of $250 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{nm}^{-2}$. A schematic representation of the employed cut-offs and distance restraint definitions is given in **Figure 3-2** below.

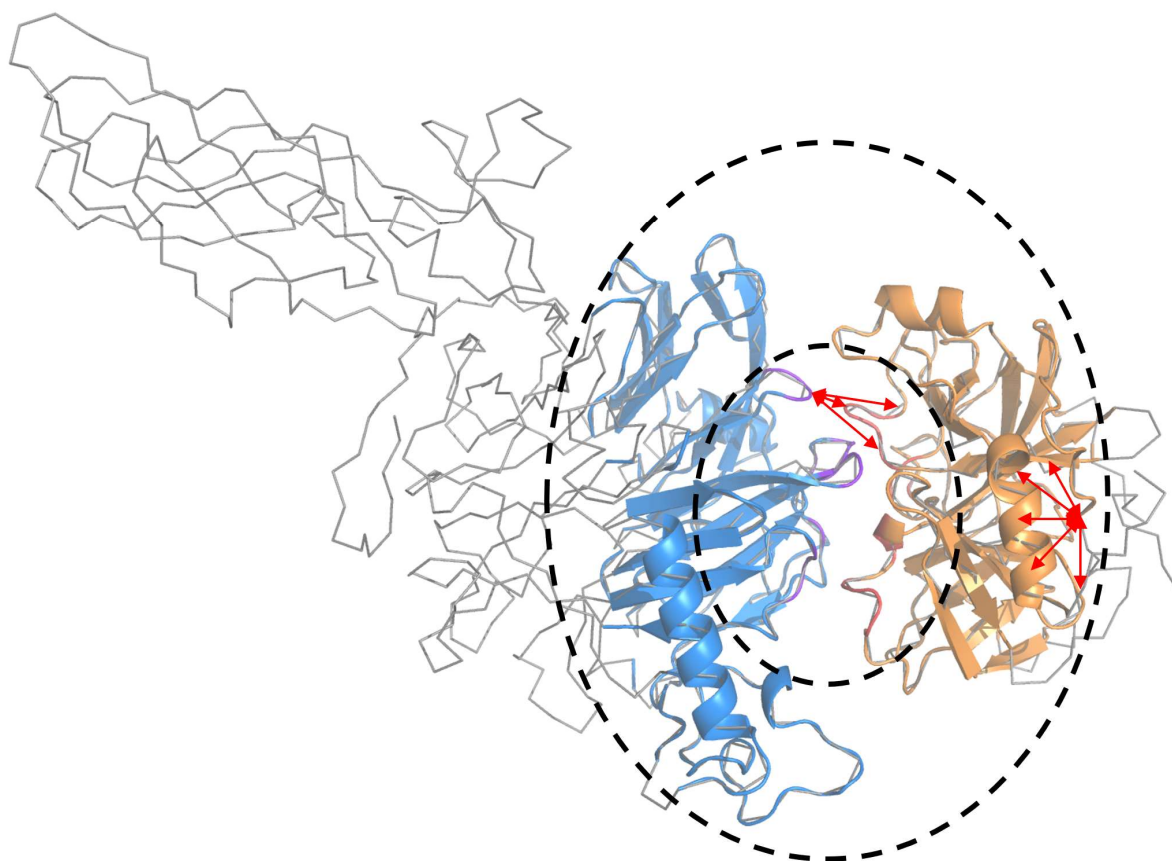


Figure 3-2. Schematic representation of employed cut-offs and distance restraints for bidirectional simulation of protein-protein binding with trimming of protein structures (example pose from CAPRI benchmark set T89). The original complex structure is shown in grey ribbon presentation, while the kept protein residues after trimming beyond a certain cut-off from the interface region (purple and red regions) are shown in blue and orange cartoon representation. The two different sets of distance restraints are shown for single atoms with red arrows: distance restraints within the interface region to pull the binding partners apart and together, and distance with constant distance to restrain the C α -atoms of artificial chain-breaks introduced by trimming.

3.3. Results & Discussion

Both additions to the GroScore pipeline – bidirectional pulling and trimming the protein structures – were tested with three different CAPRI benchmark sets (T41, T47 and T89) and compared to results obtained using the unidirectional pulling approach introduced in the previous chapter.¹ For the trimming of protein structures, three different distance cut-offs from the protein-protein interface atoms were used and compared (1.5 nm, 2.0 nm and 2.5 nm). These cut-offs were similar in length or larger than the cut-off radii for non-bonded interactions typically used in simulations employing the GROMOS force-fields,¹³ hence the most important surrounding atoms of the protein-protein interface were retained.

To assess the performance of the different methods used, the number of near-native poses ranked in the Top-10 scores and the area under the enrichment curve of scores poses (AUC) were compared. The results for all investigated protocols are given in Table 3-1 below. All bidirectional pulling methods showed increased scoring performance compared the unidirectional method, regardless of the employed trimming scheme. The strongest improvement was obtained for target T89. Here, high-quality structures were ranked in the Top-10 scored poses, while no near-native poses were obtained with the unidirectional pulling scheme. Notably, for this most recent target,

also the best performing scorer during the CAPRI challenge was not able to rank high-quality poses in their Top-10 structures.²

Table 3-1. Scoring performance of different pulling methods (unidirectional, bidirectional and bidirectional with different trimming cut-offs) as the number of structures of acceptable or better quality according to the CAPRI classification in the Top-10 ranked complexes and AUC.

Method/Performance	T41 – Top 10 (AUC)	T47 – Top 10 (AUC)	T89 – Top 10 (AUC)
Unidirectional¹	5/1** (0.68)	10/6***/4** (0.67)	0 (0.45)
Bidirectional	9/6** (0.71)	10/8***/2** (0.70)	5/4***/1** (0.54)
Bid. + 1.5 nm Cut-Off	8/3** (0.68)	10/6***/4** (0.69)	6/1***/5** (0.54)
Bid. + 2.0 nm Cut-Off	7/2** (0.71)	10/9***/1** (0.70)	7/2***/3** (0.55)
Bid. + 2.5 nm Cut-Off	8/3** (0.73)	10/6***/4** (0.71)	8/3***/5** (0.58)
Best CAPRI Scorer	10/2*** ^{18, 19}	10/10*** ^{20, 21}	10/4** ²

**medium quality

***high quality

Scatter plots of the GroScore free-energy scores against the interface-RMSDs (i-RMSDs) of the poses are given in Figure 3-3 for the whole benchmark set T89 below. Also here, it can be seen that with the bidirectional pulling protocols, more favourable GroScore free-energy scores are attributed more to near-native poses in general than with the unidirectional protocol, and the group of near-native poses is much better separated by the employed scoring schemes.

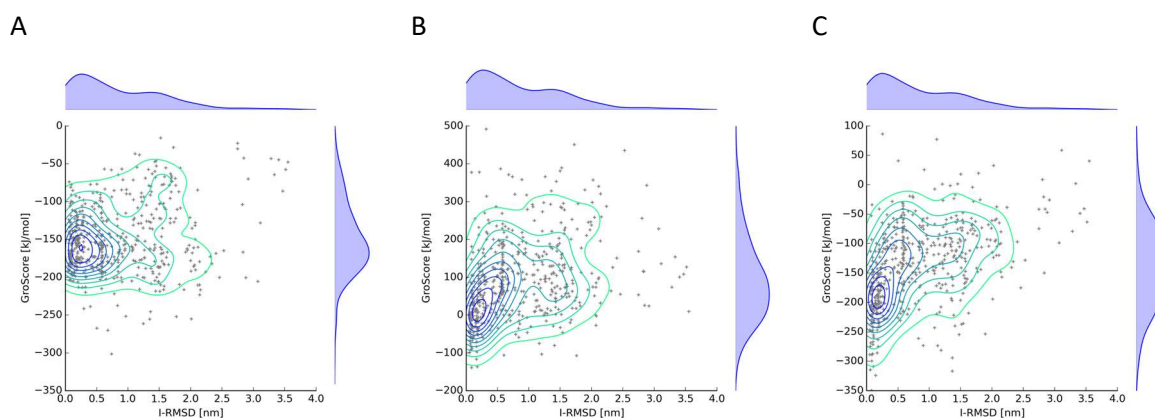


Figure 3-3. Scatter plots of calculated GroScore free-energy scores obtained with different methods against the interface-RMSDs (i-RMSD) of the docked poses with respect to the experimentally determined structures for CAPRI target T89. Panel A shows the result from unidirectional pulling (adapted from previous work),¹ panel B from bidirectional pulling and panel C from bidirectional pulling with a trimming cut-off of 2.5 nm from the interface atoms. Approximate one- and two-dimensional distributions of values shown with blue histograms and green-blue contour lines were computed using kernel density estimation.

To illustrate a possible reason for the improved scoring performance of the bidirectional pulling protocols in addition to the overcoming of the numerical bias of the exponential averaging of work values,⁵⁻⁹ the Pearson correlation coefficients of the buried surface area of the investigated poses vs. the computed GroScore is given in Table 3-2 below. With the unidirectional approach, there was a much stronger anticorrelation between the area of the protein-protein interface and the computed score, indicating a bias towards larger interface areas introduced by only pulling the proteins apart. This can be understood as an increased kinetic barrier of solvating larger interfaces. For bidirectional

simulation of unbinding and binding however, this anticorrelation is much smaller, as now also the kinetic barrier of forming a large interface is considered.

Table 3-2. Pearson correlation coefficients ρ for calculated GroScores and buried surface areas of docked complexes. Correlations are given for unidirectional and bidirectional pulling of CAPRI targets T41, T47 and T89.

Method/Correlation	T41 – ρ	T47 – ρ	T89 – ρ
Unidirectional	-0.67	-0.68	-0.73
Bidirectional	-0.48	-0.53	-0.33

Finally, the scoring performance of the different trimming schemes was compared (Figure 3-4 below). For CAPRI targets T41 and T47, the reduction in needed computational resources was only small, given the overall small size of the protein complexes. For T89 however, a large reduction in needed computational resources for scoring was achieved. Even with the largest trimming cut-off (2.5 nm), scoring needed only approximately 1/3 of computational resources compared to bidirectional pulling without trimming. Interestingly, scoring performance in terms of enrichment AUC was not reduced by trimming the protein structures. Scoring runs employing the 2.5 nm trimming cut-off even performed slightly better than scoring runs without trimming in terms of the enrichment AUC (Table 3-1 and Figure 3-4) for all investigated CAPRI targets.

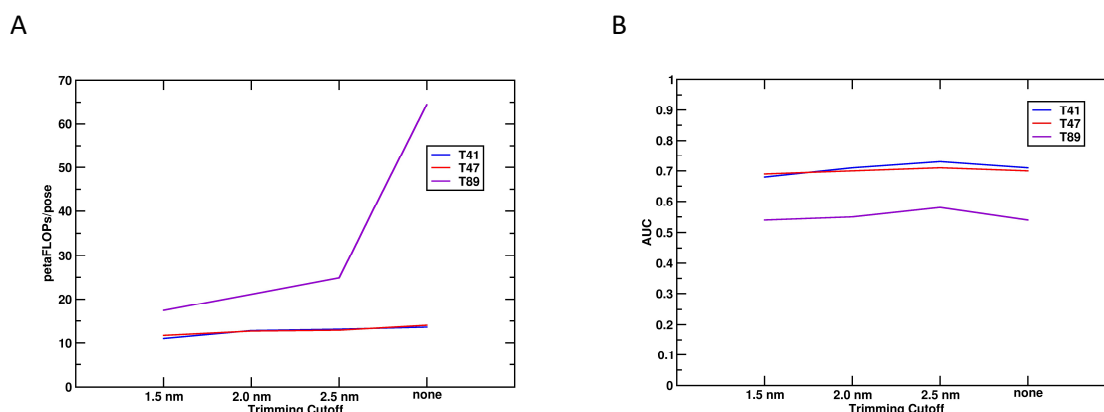


Figure 3-4. Computational resources needed (panel A) and scoring performance (panel B) for CAPRI benchmark set T89 for bidirectional pulling with different protein trimming cut-offs and no trimming.

While the effect of better scoring performance of the trimming scheme with 2.5 nm was very small, structural changes and flexibility of proteins during unbinding and re-binding might explain the observed differences. Figure 3-5 below shows the distribution of RMSDs of atom positions between the single bound and re-bound protein binding partners of CAPRI target T89. It can be observed that for 2.0 nm and 2.5 nm trimming cut-offs, the distributions look very similar, while the distribution of the 1.5 nm trimming cut-off protocol is shifted towards higher RMSDs, indicating less rigid protein structures. This is probably caused by a too stringent trimming, which retains little of the protein structures essential for stability. The distribution obtained for the whole proteins (no trimming) shows, as expected, also slightly higher RMSD values. Possibly, the protein structures obtained by trimming with 2.0 nm are still restrained slightly too much by the elastic networks applied to the introduced artificial chain termini. By contrast, the missing restraining of protein structures in the scoring protocol without trimming might allow for too strong protein deformation caused by the artificially high forces present in the non-equilibrium pulling simulations, leading to slightly reduced scoring performance. This possible “sweet-spot” of protein trimming cut-off would have to be

confirmed with further testing and could offer the possibility for further tuning of cut-off radii and restraining force-constants. Additionally, a reduction of pulling speed might alleviate the observed effects.

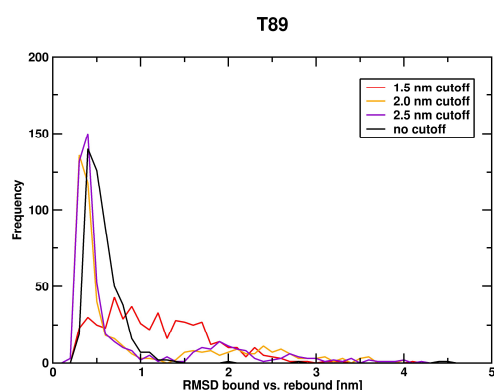


Figure 3-5. Distribution of RMSDs of atom positions between the single bound and re-bound protein binding partners for all simulated unbinding and binding reactions of CAPRI target T89 for bidirectional pulling with different protein trimming cut-offs and no trimming. To allow direct comparison, RMSDs were only computed for atoms within a 1.5 nm trimming cut-off.

3.4. Conclusion

In this chapter, two optimization strategies for the previously published GroScore scoring pipeline were introduced and tested with three different CAPRI targets. First, a bidirectional pulling approach showed increased scoring performance in terms of enrichment of near-native poses. Second, focusing simulations on only a small subset of atoms including the protein-protein interface and surrounding atoms lead to more efficient scoring as fewer computational resources were needed for similar scoring performance. Finer tuning of protein trimming cut-offs for the simulation of only the important structural parts of the protein-protein interfaces might still slightly enhance scoring performance. The presented methods still should be evaluated and tested on more scoring targets, and further assessed in blind scoring experiments.

3.5. References

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